

All Exercises with Solutions

Chapter 3: Diode Models and Circuits

Exercises 1 – 36

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1 Ideal Diode Exercises (Ex. 1 – 11)

Exercise 1 — Three-Diode Series Combinations

Problem: Determine all possible series combinations of **three** diodes and study their conduction properties.

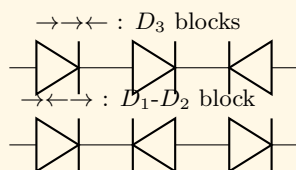
Solution

With 3 diodes, each can point $\rightarrow$ or $\leftarrow$: $2^3 = 8$ combinations.

Conducts only if ALL point the same direction:

- $\rightarrow\rightarrow\rightarrow$: Conducts A $\rightarrow$ B ✓
- $\leftarrow\leftarrow\leftarrow$: Conducts B $\rightarrow$ A ✓
- All other 6: BLOCKED (at least two neighbors oppose each other)

Key Blocked Cases



General rule: N series diodes conduct ONLY when all N face the same direction (2 out of 2^N configurations).

Exercise 2 — Diode + Series $1\ \Omega$ Resistor I/V

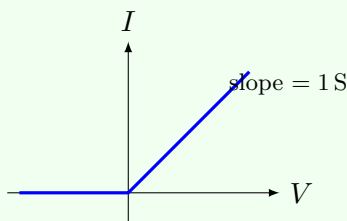
Problem: How does the I/V of Fig. 3.5(b) change if a $1\ \Omega$ resistor is placed in series?

Solution

For $V_D < 0$: No change (OFF, $I_D = 0$, no drop across R).

For $V_D > 0$: The vertical line now has slope $1/R$:

$$I_D = \frac{V - V_D}{R} = \frac{V}{1\ \Omega}$$



The perfectly vertical ON line becomes a diagonal with slope $1/R = 1\ \text{S}$. This is the transition from ideal to “slightly real.”

Exercise 3 — Battery + Parallel Diodes

Problem: Repeat Ex. 3.3 with a $1\ \text{V}$ battery in series with the parallel (antiparallel) diodes.

Solution

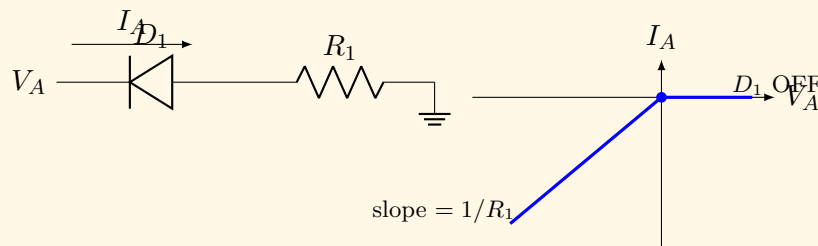
The 1V battery shifts the breakpoints:

- $V_A > 1\text{V}$: D_1 ON through the battery path
- $V_A < -1\text{V}$: D_2 ON through the battery path
- $-1 \leq V_A \leq 1$: Both OFF

The battery creates a $\pm 1\text{V}$ dead zone (no current flows for $|V_A| < 1\text{V}$).

Exercise 4 — Swapped D_1 in Diode-Resistor

Problem: Repeat Example 3.4 if D_1 terminals are swapped.

Circuit with Swapped D_1 and I/V Plot**Solution**

Now cathode faces left, anode faces right (toward ground through R_1).

- $V_A > 0$: Assume ON $\Rightarrow I_A$ flows right-to-left through D_1 (cathode $\rightarrow$ anode) $\Rightarrow I_D < 0$ **CONTRADICTION**. D_1 OFF: $I_A = 0$.
- $V_A < 0$: Assume OFF $\Rightarrow V_D = -V_A > 0$ **CONTRADICTION**. ON: $I_A = V_A/R_1 < 0$ but current flows anode $\rightarrow$ cathode (right to left, negative I_A) $\checkmark$.

$$I_A = \begin{cases} V_A/R_1 & V_A \leq 0 \\ 0 & V_A > 0 \end{cases}$$

Mirror image of Ex. 3.4 — conducts for negative instead of positive V_A .

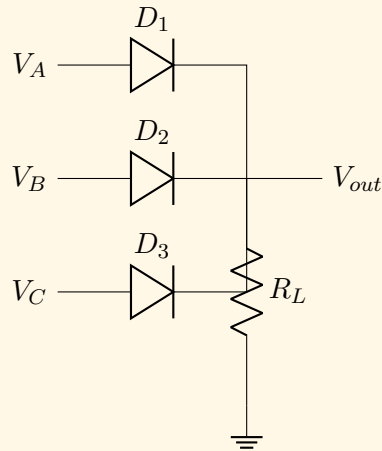
Exercise 5 — V/I Characteristic of an Ideal Diode**Solution**

The V/I plot: For $I_D > 0$: $V_D = 0$ (horizontal line on I -axis). For $I_D = 0$: $V_D \leq 0$ (horizontal line on negative V -axis).

Shape: L-shaped, rotated 90° compared to the I/V plot.

Exercise 6 — Three-Input OR Gate

Problem: Construct a three-input OR gate using diodes.

Solution**Three-Input Diode OR Gate**

$V_{out} = \max(V_A, V_B, V_C)$. Generalizes to N inputs: add N diodes.

Why it works: The highest input forward-biases its diode. All lower inputs have their diode reverse-biased (cathode at $V_{out} >$ their anode).

Exercise 7 — $1\ \Omega$ in Series at $V_D = 0$

Problem: Repeat Ex. 3.7 with $1\ \Omega$ in series with the diode.

Solution

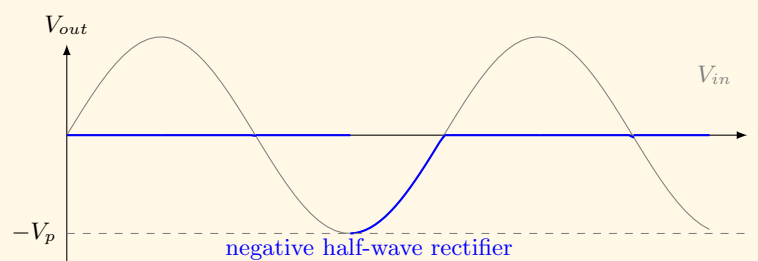
The ambiguity at $V_D = 0$ is resolved: with R in series, the system has a unique solution:

$$I = 0 \text{ and } V_D = 0$$

The resistor provides a unique I/V intersection point. No perturbation needed.

Exercise 8 — Swapped D_1 in Rectifier

Problem: Plot the characteristic of Fig. 3.10(d) if D_1 is swapped.

Swapped Rectifier Output Waveform

Solution

Swapped D_1 : now passes negative half-cycles, blocks positive.

$$V_{out} = \begin{cases} 0 & V_{in} > 0 \\ V_{in} & V_{in} \leq 0 \end{cases}$$

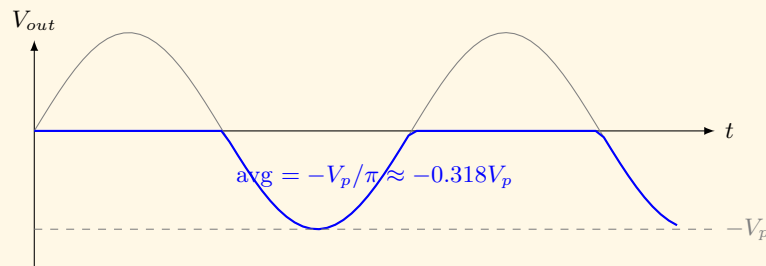
Output: only negative half of the sine wave. Average $= -V_p/\pi$.

Exercise 9 — Ex. 3.9 with $1\ \Omega$ Resistor**Solution**

Adding $1\ \Omega$ in series with the diode:

$$V_{out,avg} = \frac{V_p}{\pi} \cdot \frac{R_L}{R_L + R_s}$$

For $R_s = 1\ \Omega$ and R_L typically $\gg 1\ \Omega$: negligible effect. Reduction factor $\approx 1 - R_s/R_L$.

Exercise 10 — Swapped D_1 in HW Rectifier**Negative HW Rectifier Waveform****Solution**

Output flips: $V_{out} = V_{in}$ for $V_{in} < 0$, zero for $V_{in} > 0$. Average: $V_{out,avg} = -V_p/\pi$ (negative DC).

Exercise 11 — Pass Only Positive Cycles

Problem: How to modify Fig. 3.11(b) to pass only positive cycles?

Solution

Swap the diode direction. In 3.11(b), the diode conducts for negative V_{in} . Reversing it makes it conduct for positive V_{in} while blocking negative cycles.

2 CV Model & Divider Exercises (Ex. 12 – 18)

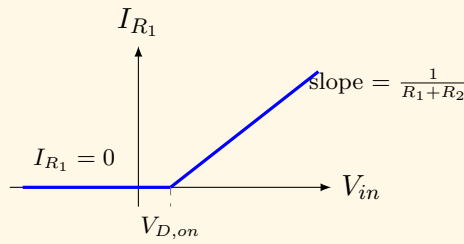
Exercise 12 — Pass Positive Cycles (Fig. 3.11b with swap)

Solution

Same as Ex. 11. Alternatively, swap the polarity of the battery V_B if one is present.

Exercise 13 — Current Through R_1 vs. V_{in} (Ex. 3.12)

I_{R_1} vs. V_{in} Graph



Solution

From Ex. 3.12 (voltage divider with diode):

$$I_{R_1} = \frac{V_{in} - V_{out}}{R_1}$$

- $V_{in} < V_{D,on}$: $V_{out} = V_{in} \Rightarrow I_{R_1} = 0$
- $V_{in} \geq V_{D,on}$: $I_{R_1} = \frac{V_{in} - V_{out}}{R_1}$ where V_{out} given by Eq. (3.11)

$$I_{R_1} = \begin{cases} 0 & V_{in} < V_{D,on} \\ \frac{V_{in} - V_{D,on}}{R_1 + R_2} & V_{in} \geq V_{D,on} \end{cases}$$

Plot: Zero until $V_{D,on}$, then linear with slope $1/(R_1 + R_2)$.

Exercise 14 — Swapped Anode/Cathode in Ex. 3.14

Solution

Swapping D_1 : now cathode connects to top, anode to ground.

D_1 ON when $V_{out} < -V_{D,on}$ (negative enough to forward-bias the flipped diode):

$$V_{out} = \begin{cases} -V_{D,on} & V_{in} < -(1 + R_1/R_2)V_{D,on} \\ \frac{R_2}{R_1 + R_2} V_{in} & V_{in} \geq -(1 + R_1/R_2)V_{D,on} \end{cases}$$

Limits the negative swing to $-V_{D,on}$.

Exercise 15 — Current Through R_1 in Ex. 3.14**Solution**

$$I_{R_1} = \frac{V_{in} - V_{out}}{R_1}$$

Same approach as Ex. 13, with the breakpoint at $V_{in} = (1 + R_1/R_2)V_{D,on}$.

Exercise 16 — Swapped Terminals in Ex. 3.15**Solution**

Mirror of Ex. 3.15: diode conducts for positive V_{in} instead of negative. Output clips at $+V_{D,on}$ for large positive V_{in} .

Exercise 17 — D_2 Before D_1 in Ex. 3.16

Problem: Assume D_2 turns on before D_1 and show contradiction.

Solution

For very negative V_{in} : D_2 ON requires current from ground through battery and D_2 to V_{in} . But D_2 's anode is at $V_B > 0$, cathode at V_{out} . With $V_{in} \ll 0$: current would flow cathode→anode in $D_1 \Rightarrow$ impossible if D_1 OFF.

If D_2 ON while D_1 OFF: $V_{out} = V_B + V_{D,on}$. But $V_{in} \ll 0$ means current flows into R_1 from the positive output, which requires I_{D_2} to flow cathode→anode $\Rightarrow$ **CONTRADICTION**. $\square$

Exercise 18 — Output Voltage of 2.35V Design

Problem: Repeat Ex. 3.17 for desired output of 2.35V.

Solution

$V_{out} = 3V_D = 2.35 \text{ V} \Rightarrow V_D = 0.783 \text{ V}$ per diode.

$$I_X = \frac{3 - 2.35}{100} = 6.5 \text{ mA}$$

$$I_S = \frac{6.5 \text{ mA}}{\exp(0.783/0.026)} = \frac{6.5 \times 10^{-3}}{1.179 \times 10^{13}} = 5.51 \times 10^{-16} \text{ A}$$

Slightly different I_S because different operating point.

3 Small-Signal Exercises (Ex. 19 – 23)**Exercise 19 — Calculating I_D from r_d**

Problem: $\Delta I_D = 1 \text{ mA}$ causes $\Delta V_D = 3 \text{ mV}$. Find I_D .

Solution

$$r_d = \frac{\Delta V_D}{\Delta I_D} = \frac{3 \text{ mV}}{1 \text{ mA}} = 3 \Omega$$

$$I_D = \frac{V_T}{r_d} = \frac{26 \text{ mV}}{3 \Omega} = \boxed{8.67 \text{ mA}}$$

Exercise 20 — Finding I_S from I_p and V_0

Problem: $I_p = 0.1 \text{ mA}$, $V_0 = 800 \text{ mV}$, $V_p = 1.5 \text{ mV}$. Find I_S .

Solution

$$I_S = \frac{I_p}{\exp(V_0/V_T)} = \frac{0.1 \times 10^{-3}}{\exp(800/26)} = \frac{10^{-4}}{\exp(30.77)}$$

$$I_S = \frac{10^{-4}}{2.34 \times 10^{13}} = \boxed{4.27 \times 10^{-18} \text{ A}}$$

Cross-check: $r_d = 26 \text{ mV}/0.1 \text{ mA} = 260 \Omega$. With $V_p = 1.5 \text{ mV}$ ($\ll V_T = 26 \text{ mV}$), small-signal valid ✓.

Exercise 21 — Derivation via dV_D/dI_D **Solution**

From $V_D = V_T \ln(I_D/I_S)$:

$$\frac{dV_D}{dI_D} = V_T \cdot \frac{1}{I_D/I_S} \cdot \frac{1}{I_S} = \frac{V_T}{I_D} = r_d \checkmark$$

Same as the forward derivation in Ex. 3.20.

Exercise 22 — Charger with $R_1 = 200 \Omega$ **Solution**

Repeat Ex. 3.17 and 3.21 with $R_1 = 200 \Omega$:

$$I_X = \frac{3 - 2.4}{200} = 3 \text{ mA}$$

$$r_d = \frac{26 \text{ mV}}{3 \text{ mA}} = 8.67 \Omega, \quad 3r_d = 26 \Omega$$

$$v_{out} = \frac{26}{226} \times 100 \text{ mV} = \boxed{11.5 \text{ mV}}$$

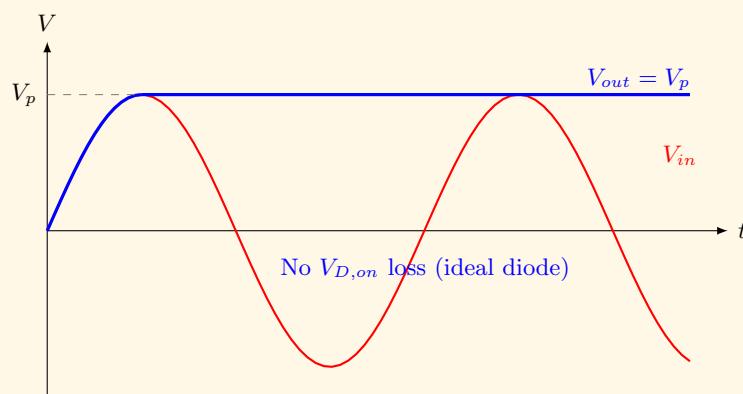
Surprisingly same! Because changing R_1 changes both the bias current and the divider ratio, and they partially cancel.

Exercise 23 — $R_1 = 80\ \Omega$ **Solution**

$$I_X = \frac{3 - 2.4}{80} = 7.5\ \text{mA}, \quad r_d = \frac{26}{7.5} = 3.47\ \Omega, \quad 3r_d = 10.4\ \Omega$$

$$v_{out} = \frac{10.4}{90.4} \times 100 = \boxed{11.5\ \text{mV}}$$

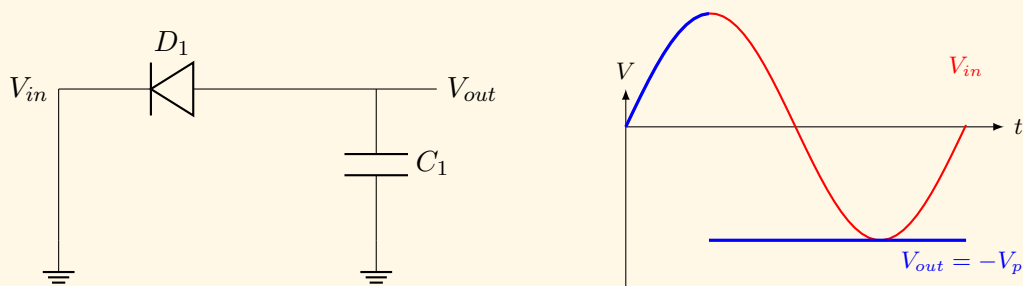
Again 11.5 mV! The product $R_1 \times I_X$ remains constant $\Rightarrow 3r_d/(R_1 + 3r_d)$ is approximately $3V_T/(V_{in} - V_{out}) = 78/(600) = 0.13$ regardless of R_1 .

4 Rectifier & Capacitor Exercises (Ex. 24 – 29)**Exercise 24** — HW Rectifier with Ideal D_1 **Ideal Peak Detector Waveform****Solution**

With ideal D_1 : $V_{D,on} = 0$. Peak detector charges to V_p exactly.

$$V_{out} = V_p \quad (\text{constant, no load})$$

Ripple with load: $V_R = V_p/(R_L C_1 f)$ (no $V_{D,on}$ subtracted).

Exercise 25 — Diode Swapped in Peak Detector**Negative Peak Detector Circuit and Waveform**

Solution

Reversed D_1 : charges C_1 during **negative** peaks. $V_{out} = -(V_p - V_{D,on})$ (constant negative DC output). This is a **negative peak detector**. Used in precision rectification circuits.

Exercise 26 — $1\ \Omega$ Load Power from 3.3V**Solution**

$$P = \frac{V^2}{R} = \frac{(3.3)^2}{1} = \boxed{10.89\ \text{W}}$$

Or equivalently: $I = V/R = 3.3\ \text{A}$, $P = IV = 3.3 \times 3.3 = 10.89\ \text{W}$.

Exercise 27 — Varying R_L with Fixed C_1 **Solution**

From $V_R = (V_p - V_{D,on})/(R_L C_1 f)$:

Increasing $R_L \Rightarrow$ decreasing V_R (less load current, slower discharge).

R_L	I_L	V_R
$0.5\ \Omega$	$7.4\ \text{A}$	$\sim 2V_p$ (useless)
$10\ \Omega$	$0.37\ \text{A}$	moderate
∞	0	0 (ideal DC)

Exercise 28 — 220V/50Hz Line**Solution**

$$V_p = 220\sqrt{2} = 311\ \text{V}, f = 50\ \text{Hz}.$$

Design comparison:

$$C_1 = \frac{V_p - V_{D,on}}{V_R \cdot R_L \cdot f}$$

At 50 Hz vs. 60 Hz: need $60/50 = 1.2\times$ more capacitance for same ripple.

Which frequency gives better C_1 ? 60 Hz gives smaller C_1 (faster recharge cycles).

Exercise 29 — $C_1 = 1000\ \mu\text{F}$ **Solution**

Using Ex. 3.28 parameters with $C_1 = 1000\ \mu\text{F} = 0.001\ \text{F}$:

$$V_R = \frac{V_p - V_{D,on}}{R_L \cdot C_1 \cdot f}$$

Substitute specific values from the example to get V_R . A $1000\ \mu\text{F}$ capacitor is practical for moderate loads but gives large ripple for heavy (low- R_L) loads.

5 Regulator & Limiter Exercises (Ex. 30 – 36)

Exercise 30 — Zener Operating Conditions

Solution

For the Zener to regulate:

1. $V_{in} > V_Z + V_{D,on}$ (enough headroom to bias the Zener into breakdown)
2. $I_Z > I_{Z,min}$ (minimum current for stable breakdown)
3. R_1 chosen so: $I_{Z,min} < \frac{V_{in}-V_Z}{R_1} < I_{Z,max}$

If V_{in} too low: Zener comes out of breakdown $\Rightarrow$ no regulation.

Exercise 31 — Zener with Varying Input

Solution

V_{in} varies ± 1 V around nominal. Using line regulation:

$$\Delta V_{out} = \frac{r_D}{r_D + R_1} \Delta V_{in}$$

With $r_D = 5 \Omega$, $R_1 = 1 \text{ k}\Omega$:

$$\Delta V_{out} = \frac{5}{1005} \times 2 \text{ V} = \boxed{10 \text{ mV}}$$

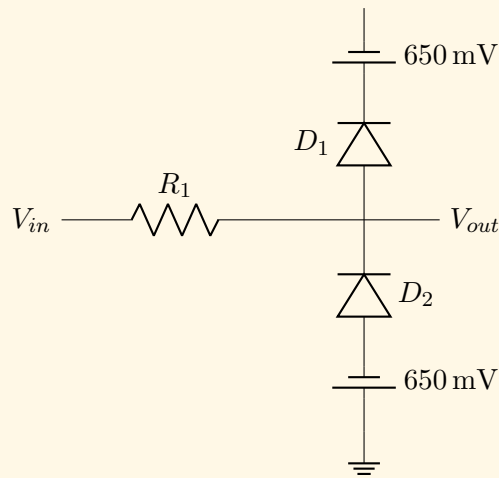
2V input swing produces only 10 mV output change — excellent regulation.

Exercise 32 — Limiter Design Problem

Problem: Design a limiter at $\pm 150 \text{ mV}$ with $V_{D,on} = 800 \text{ mV}$.

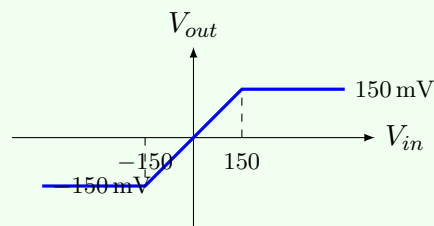
Solution

$$V_B = V_L - V_{D,on} = 150 - 800 = \boxed{-650 \text{ mV}}$$

Circuit

Negative batteries **oppose** the diode drops, pulling the effective clamping level from 800 mV inward to 150 mV.

Transfer characteristic:

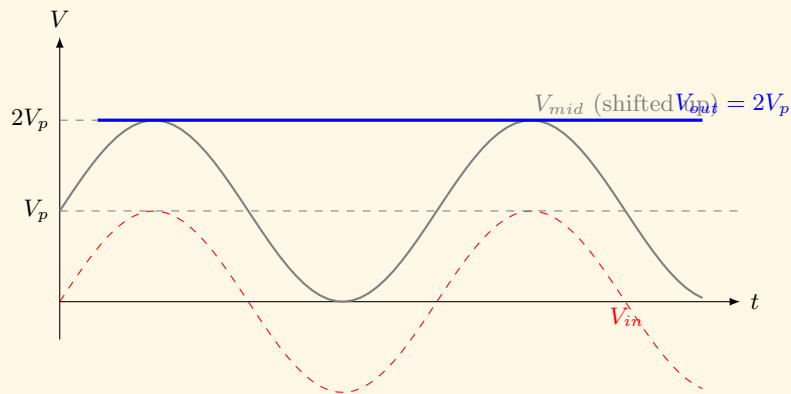
**Exercise 33 — Capacitive Divider Analysis****Solution**

For capacitors in series with a step input ΔV_{in} :

$$\Delta V_{out} = \frac{C_1}{C_1 + C_2} \Delta V_{in}$$

Key difference from resistive divider: C_1 (input cap) is in the **numerator**, not C_2 . Smaller $C_2 \Rightarrow$ **larger** output (opposite of resistors).

Why? Charge is conserved. If C_2 is small, a small amount of charge produces a large voltage change across it.

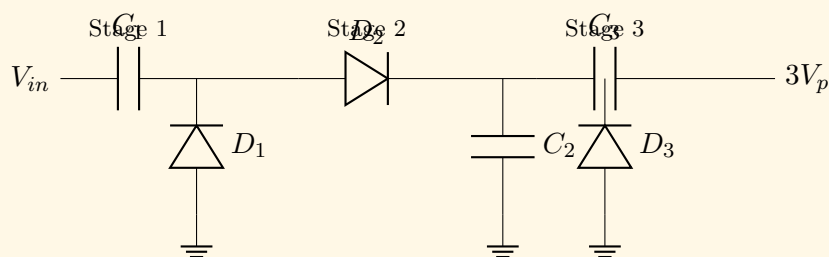
Exercise 34 — Voltage Doubler Operation**Voltage Doubler Waveforms****Solution****Stage 1 (Clamper):**

- Negative swing: D_1 ON, charges C_1 to V_p
- Result: signal is DC-shifted up by V_p
- V_{mid} swings from 0 to $2V_p$

Stage 2 (Peak Detector):

- D_2 ON at peak of V_{mid} , charges C_2 to $2V_p$
- $V_{out} = 2V_p$

With losses: $V_{out} = 2V_p - 2V_{D,on}$ (one diode drop per stage).

Exercise 35 — Voltage Tripler**Voltage Tripler Circuit****Solution**

Add a third C-D stage after the doubler:

- Stage 3 clamper shifts the $2V_p$ signal by another V_p
- Peak detector captures $3V_p$

$$V_{out} = 3V_p - 3V_{D,on}$$

General rule: N stages $\Rightarrow V_{out} = NV_p - NV_{D,on}$.

Exercise 36 — Full-Wave vs. Half-Wave Comparison**Solution**

Parameter	Half-Wave	Full-Wave Bridge
Number of diodes	1	4
Drop per cycle	$V_{D,on}$	$2V_{D,on}$
Ripple formula	$\frac{V_p - V_{D,on}}{R_L C_1 f}$	$\frac{V_p - 2V_{D,on}}{2R_L C_1 f}$
Effective ripple freq.	f	$2f$
Peak reverse voltage	$2V_p$	V_p
Avg output (no cap)	V_p/π	$2V_p/\pi$

Conclusion: FW is better in every way except cost (4 diodes, $2V_{D,on}$ loss). For most power supplies, FW is preferred because the ripple is **halved** and reverse voltage stress is halved.

Good luck on your quiz!

Remember: The diode is just a one-way valve.
Everything else is KVL, KCL, and careful bookkeeping.